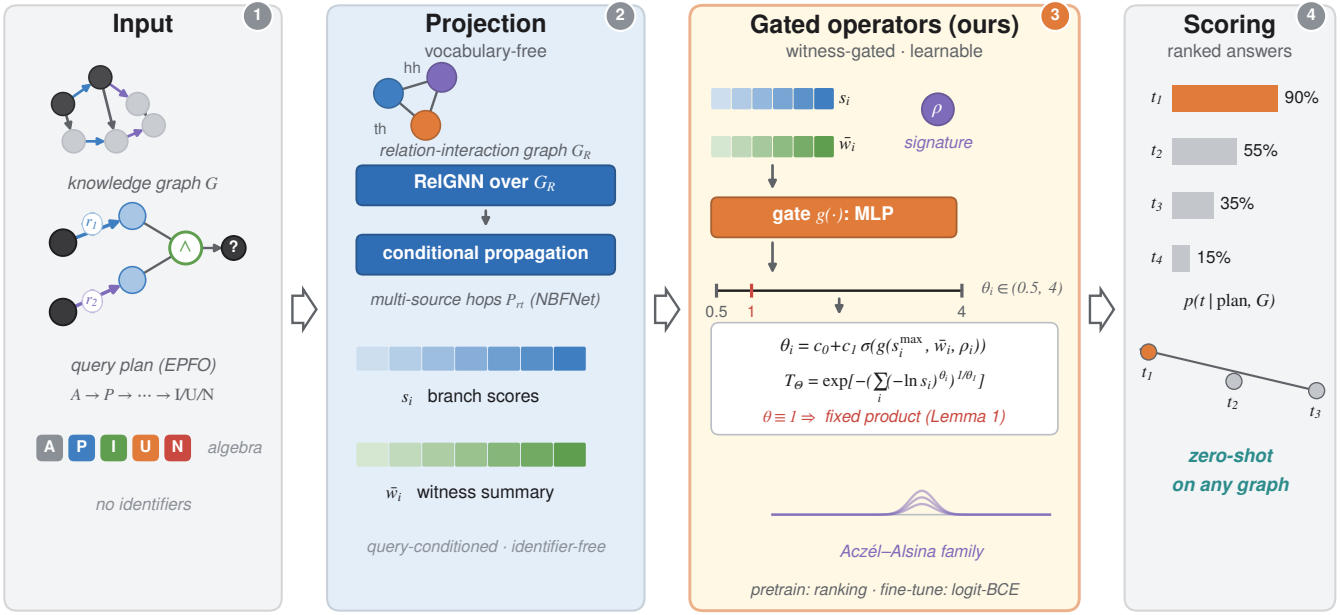
Input
1
knowledge graph G
r_1
r_2
∧
?
query plan (EPFO)
$A \to P \to \cdots \to I/U/N$
A P I U N
algebra
no identifiers
Projection
2
vocabulary-free
hh
th
relation-interaction graph G_R
RelGNN over G_R
conditional propagation
multi-source hops P_{rt} (NBFNet)
s_i branch scores
$\bar{w}_i$ witness summary
query-conditioned · identifier-free
Gated operators (ours)
3
witness-gated · learnable
s_i
$\bar{w}_i$
ρ
signature
gate $g(\cdot)$: MLP
$\theta_i \in (0.5,\ 4)$
0.5
1
4
$\theta_i = c_0 + c_1\,\sigma(g(s_i^{\max}, \bar{w}_i, \rho_i))$
$T_\Theta = \exp[-(\sum_i [-\ln s_i)^{\theta_i}]^{1/\theta_i}]$
$\theta \equiv 1 \Rightarrow$ fixed product (Lemma 1)
Aczél–Alsina family
pretrain: ranking · fine-tune: logit-BCE
Scoring
4
ranked answers
t_1 90%
t_2 55%
t_3 35%
t_4 15%
$p(t \mid \text{plan}, G)$
t_1
t_2
t_3
zero-shot
on any graph